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CELLULAR MESSENGERS AND RECEPTORS

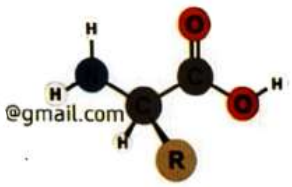
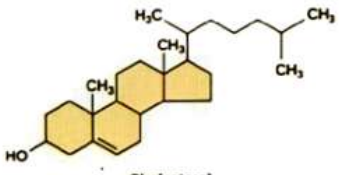
Hormones act on cells by receptors.

Receptors can be :

- Extracellular.
- Intracellular.

Classification of Hormones & Hormone Receptors 00:01:37

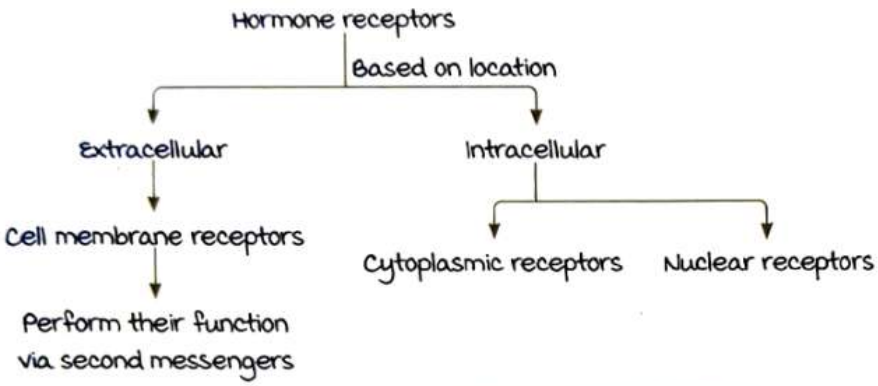
CLASSIFICATION OF HORMONES :

Class	Types	Features
Amino acid derivatives  @gmail.com	• Tyrosine : – Epinephrine. – Norepinephrine. – Dopamine. } Catecholamines – Thyroid hormones. • Tryptophan : – Serotonin. – Melatonin. • Histidine : Histamine.	Cannot cross the cell membrane → Extracellular receptors
Peptide & protein hormones 	• 3 AA's : Thyrotropin releasing hormone (TRH). • 51 AA's : Insulin. • 84 AA's : Parathormone.	Cannot cross the cell membrane → Extracellular receptors
Cholesterol derivatives  Cholesterol	Steroid group of hormones : • Estrogen. • Testosterone. • Aldosterone. • Cortisol.	Cross cell membrane → Intracellular receptor
Vitamins  Cholecalciferol Vitamin D_3	• Vitamin A. • Vitamin D.	Lipid soluble → Easily crosses cell membrane → Intracellular receptor

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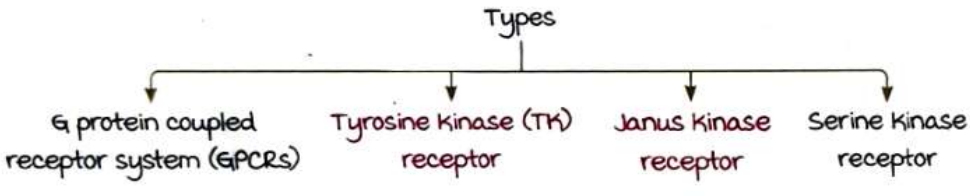
CLASSIFICATION OF HORMONE RECEPTORS :

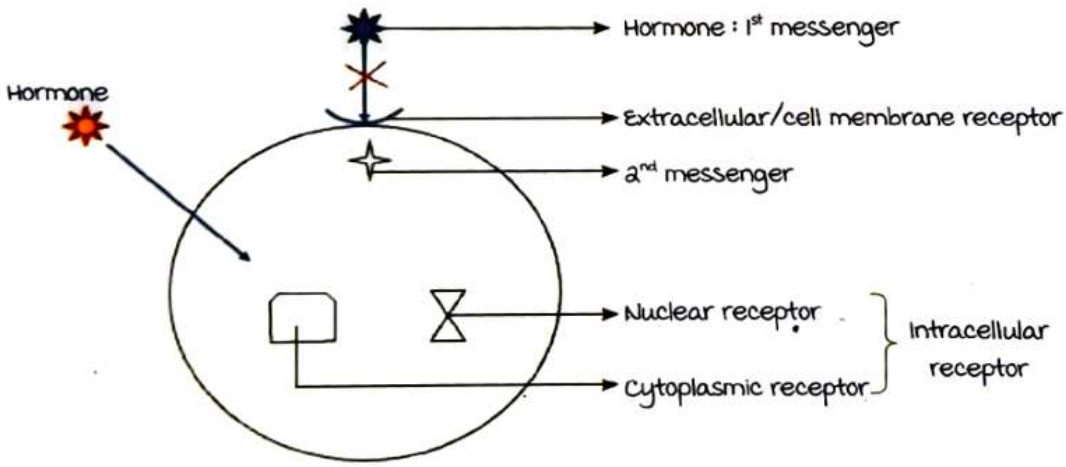
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Extracellular Receptors

00:08:40

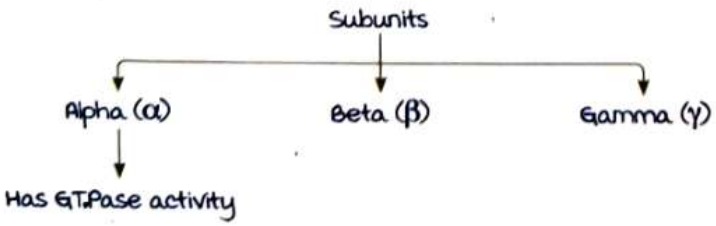




G-PROTEIN COUPLED RECEPTOR :

- Transmembrane protein.
- AKA serpentine receptors/7 transmembrane receptors.

Subunits of the receptor :



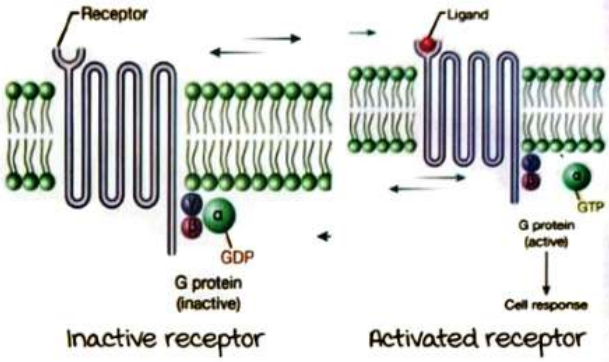
Feedback

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mechanism of action :

Hormone binds to GPCR
↓
α-subunit gets seperated
↓
Hydrolyses GTP to GDP
↓
Energy liberated
↓
2nd messengers are sent inside the cell



Inactive receptor | Activated receptor

2nd messengers :

- Enter the cell & influence cellular function.
- Types :
  - Cyclic AMP (c-AMP)
  - Ca^{2+} (IP3-DAG-Ca^{2+})
  - Cyclic GMP (c-GMP)

} Activate protein kinases → Regulate downstream reactions.

GPCR can use any 2nd messenger system :

a. G_s (Stimulatory) → stimulates adenylyl cyclase → ↑ cAMP.
b. G_i (Inhibitory) → inhibits adenylyl cyclase → ↓ cAMP.
c. G_q → ↑ Ca^{2+}.
d. G_t → acts by transducin in retinal rods → activates PDE enzymes → ↓ cGMP.

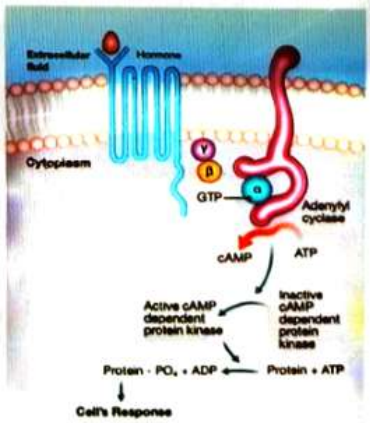
Cyclic AMP :

Effects of cAMP (2nd messenger)

metabolism
- Glycogenolysis
  - Glucagon & epinephrine (1st messenger)

Electrolyte secretion
Secretin (1st messenger)
↓ cAMP
↑ HCO_3^- in pancreatic juice.
HCO_3^- neutralizes acid.
Secretin is nature's antacid.

Water reabsorption
Vasopressin (V_2) (1st messenger)
↓ cAMP
↑ Aquaporin-2



Activation of cAMP

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Clinical application of electrolyte secretion :

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1. Bacterial toxins :

Cholera toxin → Stimulates G_s → ↑ R_s → ↑↑ cAMP (Pathological) → ↑ electrolyte secretion → Causes diarrhea → Dehydration, death

Pertussis toxin → Inhibits G_i → ↑↑ cAMP (Pathological) → Electrolyte secretion in sputum → Whooping cough

2. Gain of function mutation of $G_{s\alpha}$ subunit :

↑↑ cAMP

- Precocious puberty
- Skeletal malformations
- Cafe au lait spots

} McCune Albright syndrome

Calcium :

Effects of Ca^{2+} 2nd messenger

- Uterine contraction — Oxytocin (1st messenger)
- Vasoconstriction — Vasopressin (V1 receptor), Angiotensin } 1st messengers
- Exocytosis — TRH, Cholecystokinin (CCK) } 1st messenger

Effects of cGMP (Opposite of Ca^{2+})

Relaxation & vasodilatation

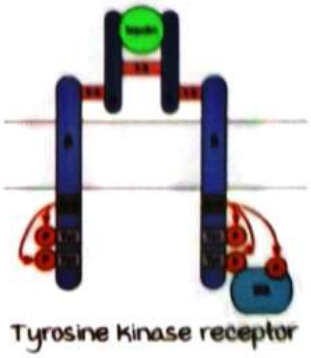
By nitric oxide & natriuretic peptide (e.g : Atrial natriuretic peptide)

Tyrosine kinase receptor :

Insulin → TK receptor → ↑ cell growth & differentiation.

Clinical application :

BCR-ABL translocation (Philadelphia chromosome) → Activates tyrosine kinase → Pathological cell growth & differentiation → CML



Tyrosine kinase receptor

Rx of CML : Imatinib mesylate (Blocks TK receptor).

Feedback

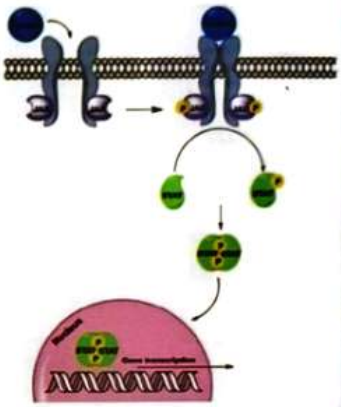
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Janus kinase receptor :

- AKA cytokine receptor.
- JAK-STAT (Signal transduction & activators of transcription).
- Acted upon by :
  - Growth hormone.
  - Prolactin.
  - Erythropoietin.
  - Leptin.



Janus Kinase receptor

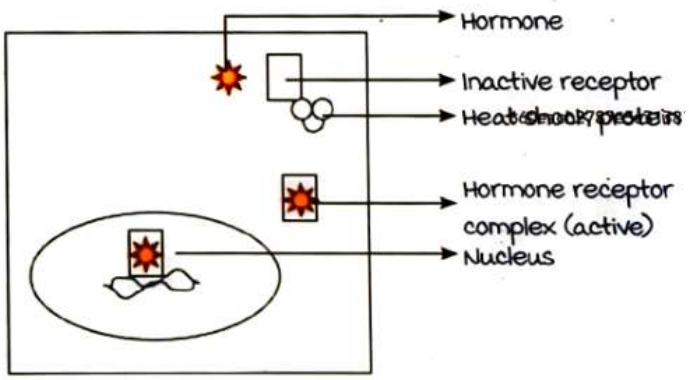
Serine kinase receptor :

- Important for reproductive functions.
- Acted upon by :
  - Activins.
  - Anti mullerian hormone.

Intracellular Receptors 00:37:31

Hormones that enter the cell act on intracellular receptors.

CYTOPLASMIC RECEPTORS :



NUCLEAR RECEPTOR :



Structure of nuclear receptor

a. A/B :
  - Near N terminal.
  - Contain activating factors (AF-1).
  - Activate transcription.

b. C : DNA binding domain.

c. D : Hinge domain.

d. E : Ligand binding domain (Eg : steroid hormones).

d. F :
  - Contains AF-2.
  - Activates transcription.

Cytoplasmic receptors	Nuclear receptors
• Mineralocorticoid receptor (Aldosterone) • Glucocorticoid receptor (Cortisol) • Androgen receptor (Testosterone) • Vitamin D receptor	• Thyroid hormone receptor • Vitamin A receptor • Estrogen receptor • Progesterone receptor

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Homeostasis of Hormones & Neurotransmitter Receptors

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SAFETY MECHANISM FOR HORMONE OVERACTIVITY :

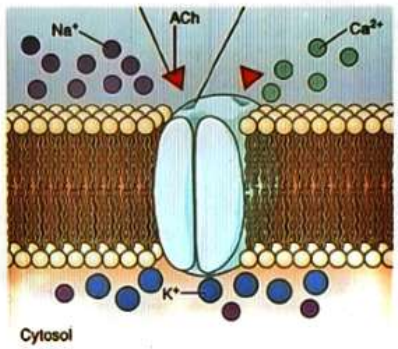
1. Internalization of receptors.
2. Desensitization of receptors : Receptors are modified to be less responsive.
3. Inactivation of receptors.

Neurotransmitter receptors :

Types :

- Ligand gated receptors :
  - Activates ion channels → Fast responses.
  - Eg : Nicotinic Ach receptors, $GABA_A$, $GABA_C$.

- GPCR/metabotropic receptors :
  - Act by 2nd messengers → Slow response.
  - Eg : Muscarinic acetylcholine receptor, $GABA_B$.



Ligand gated receptor

Feedback